

Triasha Sarkar

Machine Learning Engineer

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PROFILE SUMMARY

Machine Learning Engineer with 2+ years of aerospace industry and applied research experience building Generative AI, RAG, agentic workflows, multimodal evaluation, and production-oriented ML systems. Hands-on experience with LangGraph/LangChain orchestration, PEFT/LoRA, TRL/GRPO experimentation, hybrid retrieval, LLM evaluation, FastAPI microservices, Docker, observability, CI/CD, and reliability testing; strong Python background spanning model development, failure analysis, cloud data pipelines, and engineering-reviewed validation.

WORK EXPERIENCE

Graduate Research Assistant, Aerospace Systems Design Laboratory, Georgia Tech May 2025 - Aug 2026 Atlanta, GA

- M.S. Thesis - built a leakage-resistant failure-detection workflow for time-series thrust simulations; audited 15,120 cases, identified 6,804 candidate simulator failures, removed duplicate-geometry leakage, and achieved 96.8% accuracy / 0.986 ROC AUC on unseen geometries, replicated on an independent 1,050-case study.
- HERO - Delta Air Lines-sponsored applied ML: helped build a source-aware RAG QA workflow over FOQA/ASAP records, separating operational and safety data to support traceable retrieval for analyst-led safety and risk assessment.
- GREEN TEA - U.S. Endowment-sponsored project: built a surrogate of Argonne National Laboratory's GREET life-cycle model and integrated it with economic and transportation models in a Tableau decision-support workflow delivered to the sponsor team.

Machine Learning Engineer, Alten India, embedded at Rolls-Royce Oct 2023 - Apr 2025 Bangalore, India

- Developed Python ML workflows for aircraft-engine health using multivariate sensor diagnostics, anomaly detection, and predictive maintenance; engineered 1 Hz/10 Hz time-series features to identify failure symptoms and component-health risk.
- Built Azure Databricks pipelines over JSON, Parquet, HDF5, and CSV data in Azure Blob Storage, automating extraction, quality control, and preprocessing for downstream ML analysis.
- Partnered with lifecycle engineers to define failure modes, validate diagnostic signals against engineering knowledge, and translate outputs into pre-emptive health reports supporting proactive maintenance planning and an estimated 5% maintenance-cost reduction.

Data Science Intern, Rolls-Royce DataLabs May 2021 - Jul 2021 Bangalore, India

- Built proof-of-concept aerospace ML analyses spanning EDA, feature engineering, anomaly detection, degradation analysis, time-series/sensor-fusion workflows, and physics-informed methods with data engineers and domain experts.

PROJECTS

AeroRAG-X, Generative AI, Agentic RAG & ML Systems Oct 2025 - Present [GitHub](#)

- Architected an evidence-grounded RAG and agentic AI system over 3,233 citation-preserving NASA NTRS chunks using BM25 + MiniLM dense retrieval, RRF, cross-encoder reranking, pgvector, evidence-sufficiency gating, and application-controlled citations; extended it into a stateful LangGraph agent with schema-constrained planning, dynamic tool routing, bounded execution, grounded generation, citation validation, and inspectable trajectories.
- Engineered autonomous-agent reliability controls with immutable checkpoints, deterministic resume semantics, HITL approve/reject/edit workflows, bounded per-tool retries, structured failure classification, and fault injection for timeout, malformed-response, service-error, and dependency-unavailable scenarios so unrecoverable failures terminate safely rather than generate unsupported answers.
- Fine-tuned Qwen3-0.6B with PEFT/LoRA and built protected LLM/agent evaluation across answerability, grounded refusal, citation validity, evidence support, tool selection, and trajectory efficiency; developed a TRL/GRPO grounded-agent experiment harness with multi-objective rewards, anti-reward-hacking gates, train/evaluation leakage controls, and paired Base-vs-LoRA-vs-GRPO held-out ablation contracts.
- Decomposed the GenAI stack into independently containerized Agent, Retrieval, and Inference FastAPI services with Docker Compose and PostgreSQL/pgvector; added typed async contracts, request/trace/thread propagation, OpenTelemetry tracing, Prometheus metrics, bounded retries, health/readiness checks, cross-service citation validation, and safe degradation under downstream failure.

AeroSynth-Eval, Multimodal AI Evaluation & VLM Reliability Aug 2026 [GitHub](#)

- Designed a reproducible VLM evaluation benchmark with a frozen 48-scenario matrix, protected 36/12 development-test split, 48 deterministic synthetic assets, SHA-256 provenance, typed Pydantic/Typer evaluation contracts, strict response/schema validation, and a local MLX-VLM adapter that binds outputs to exact asset/scenario/rubric metadata.

- Automated reliability checks with GitHub Actions across Ruff, strict mypy, pytest, corpus integrity, annotation contracts, prompt/response validation, and no-inference smoke plans; malformed outputs, provenance mismatches, and protected-test access are rejected before evaluation.

NewsLens, Recommender Systems & Production ML GitHub

Jul 2026

- Built a leakage-safe MIND-small recommendation pipeline with DuckDB, chronological splits, TF-IDF personalization, and training-only popularity fallback (NDCG@10 0.3664; Recall@10 0.5955), then productionized versioned artifacts behind FastAPI, non-root Docker/Compose, telemetry, multi-arch GHCR CI/CD, and 400+ automated tests.

EdgeGenBench, Scientific ML & Efficient Inference GitHub

Aug 2026

- Built an uncertainty-aware aircraft-design benchmark with surrogate models, conformal/safety gating, constrained optimization, and physics validation; best surrogate reached 0.995 mean test R2, and ONNX export reduced batch-1 surrogate latency from 13.8 ms to 0.305 ms while preserving numerical-equivalence checks.

TECHNICAL SKILLS

Languages & data: Python, SQL/MySQL, C++ (working), MATLAB; pandas, NumPy, SciPy, DuckDB, PostgreSQL/pgvector; Azure Blob Storage, Azure Databricks, Git

GenAI & ML: Hugging Face Transformers, PEFT/LoRA, TRL/GRPO, LangGraph, LangChain, RAG, BM25, dense retrieval, RRF, cross-encoder reranking, Sentence Transformers, PyTorch, TensorFlow, scikit-learn, MLX-LM/MLX-VLM, ONNX Runtime

Systems & evaluation: FastAPI, REST APIs, Docker/Compose, Google Cloud Run, GitHub Actions, Prometheus, OpenTelemetry, CI/CD, tracing, health/readiness checks, fault injection, held-out evaluation, schema/provenance validation, failure analysis

EDUCATION

Georgia Institute of Technology, M.S., Aerospace Engineering

Aug 2026

Cranfield University, M.Sc., Aerospace Vehicle Design

Feb 2023

SRM Institute of Science and Technology, B.Tech., Aerospace Engineering

Jun 2021

LEADERSHIP

Student Chair, Women of Aerospace & Aeronautics (WOAA), Georgia Tech Chapter

- Led graduate-student communications and community engagement, coordinating recurring peer networking and chapter outreach.